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[Portfolio](#) | [LinkedIn](#) | [Github](#) | [LeetCode](#)

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Reply,
Manchester, United Kingdom

Dear Hiring Manager,

I am writing to express my interest in the Graduate AI and Software Engineer positions at Cortex Reply. With an MSc in Advanced Computer Science (Distinction) from the University of Liverpool (Russell Group, November 2024), more than ten years in software development and over three years of industry experience as a Software Engineer, I bring a solid background in software solutions, and team management. I am eager to apply my acquired knowledge and skills to contribute to the organization.

As a student, I engaged with modules such as Research Methods in Computer Science, Advanced Algorithmic Techniques, and Optimization, which deepened my understanding of AI algorithms and their optimization. I have developed strong proficiency in many programming languages, frameworks and technologies through assignments and am skilled in using the JetBrains IntelliJ IDEA and Visual Studio Code IDEs. For my MSc dissertation, I built a full-fledged auction website using React, Node.js and MySQL. I received an A grade and a distinction score for this work (a video demo is in my portfolio, [Frontend Code](#), [Backend Code](#)).

During my tenure at Deerwalk Services Pvt. Ltd. (now Cedar Gate Technologies), Kathmandu, I worked on scalable Java-based SaaS healthcare analytics platforms while collaborating closely with global teams, developers and managers across geographies was a regular part of my role. I gained hands-on experience across the full software development lifecycle (SDLC), including database design, big data visualization in Grails (Java), and working with Elasticsearch and MySQL. I contributed to major products such as Plan Analytics, Benecore KIP Stop Loss, and was involved in code refactoring to improve performance, maintainability, and testability. I actively worked in agile environments—participating in daily stand-ups, sprint planning, and iterative delivery—while using GitHub with dedicated feature branches for workflows. I thrived in a collaborative team setting, often taking initiative and occasionally leading small modules, which strengthened my communication, teamwork, and time-management skills.

I have further explored API-driven, diverse range of applications using core Java 8+ (also Spring Boot), JavaScript and other programming languages and frameworks. I have also pursued multiple certifications that underscore my commitment to continuous learning and growth:

- [Java Spring Boot 3, Apache Kafka Bootcamp-With AI Assistant](#) (Ongoing).
- [Modern Event-Driven Spring Boot: Kafka & Spring Cloud Stream](#) (Apr, 2026): [Movies](#)
- [Reactive Redis Masterclass For Java Spring Boot Developers](#)(Apr, 2026): [Caching](#), [Performance](#).
- [Cloud-Native: Microservices, Kubernetes, Service Mesh, CI/CD](#) (Mar, 2026).
- [Master Microservices with SpringBoot, Docker, Kubernetes](#) (Mar, 2026): [JaneBank](#) (9 branches).
- [From Java Dev to AI Engineer: Spring AI](#) (Jan, 2026): [SpringAI](#)([All](#), [Observability](#), [Testing](#)), [MCP Servers](#)([STDIO](#), [SSE](#)), [MCP Client](#).
- [Spring Boot Microservices and Spring Cloud. Build & Deploy](#) (Dec 2025): [Eureka Servers Cluster](#)
- [Master AI, Large Language Models & Agents](#) (Nov 2025). I worked with frontier and open-source LLMs, experimented with model manipulation, and understood how modern GenAI systems function. Throughout the course, I continuously applied concepts like RAG, model fine-tuning, and agent workflows to create practical AI tools and applications.
- [Ultimate Kubernetes Fast-Track Beginner to Advanced](#) (Nov 2025): [YAML files](#)
- [Ultimate Docker Fast-Track Beginner to Advanced](#) (Oct 2025): [Dockerfiles](#)
- [Full Stack: React and Java Spring Boot - The Developer Guide](#) (Oct 2025): [Frontend Code](#) (React in TypeScript), [Backend Code](#)
- [OAuth 2.0 in Spring Boot Applications](#) (Sep 2025): [Discovery Service using Eureka](#) with [API Gateway](#) & [Resource Server](#), [Social Login](#).
- [Spring Boot REST APIs: CRUD Operations, JWTs, OpenAPI, Swagger, MySQL](#) (Sep 2025): [TODOs Rest API](#) , [Employees Rest API](#).
- [Complete Guide to Elasticsearch](#) (Aug 2025, NoSQL).
- [Testing Java: JUnit 5, Mockito, Testcontainers](#) (Jul 2025): [JUnit5 tests](#), [TDD](#), [with testcontainers \(in Docker\)](#), [REST Assured](#)).
- [Apache Kafka for Event-Driven Spring Boot Microservices](#) (Jul 2025): [Saga Pattern Code](#).
- [Full Stack: Angular and Java Spring Boot E-Commerce Website](#) (Apr 2025). [Frontend code](#) (Angular in TypeScript), [Backend code](#) .
- [Generative AI with Java and Spring Boot | Spring AI, Local LLMs via Ollama & LM Studio - The Practical Guide, AI Agentic Design Patterns with Ollama & OpenAI Guide, Mastering Spring AI : Build AI with Java, Build AI Apps with Spring AI, OpenAI, Ollama & SpringBoot](#) (Upcoming).

I also actively hone my problem-solving skills through daily practice on LeetCode having solved over 725 problems in Java and 640 in JavaScript. This consistent practice has deepened my understanding of data structures, algorithms, and design patterns— enabling me to write more efficient, scalable, and maintainable code. Together, they demonstrate my strong technical foundation and proactive approach to growth and innovation.

I believe Cortex Reply is a great fit for me because of its strong focus on cutting-edge technologies like AI, cloud computing, and large-scale system design—areas I have already begun exploring through my academic and project work. I'm particularly drawn to the opportunity to work across AI/ML, cloud, and software engineering while collaborating with experienced consultants in diverse industries. I'm a quick learner, and I'm confident I can rapidly adapt to new tools, technologies, and environments while contributing effectively to project teams.

Yours sincerely,
Ritu Bafna